

Simulating and Optimizing Patient Flow using Discrete Event Simulation

Numerical Analysis, Simulation and Modeling Sessional

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Base Paper Overview

Paper Title: *Discrete Event Simulation-Based Analysis and Optimization of Emergency Patient Scheduling Strategies* (Lv et al., 2025)

Domain: Healthcare Management & Operations Research

Brief Summary:

- ▶ Emergency Departments (EDs) face increasing crowding and complexity.
- ▶ Traditional mathematical queuing models fail to capture this complexity due to "state space explosion."
- ▶ The paper constructs a decision support framework using Discrete Event Simulation (DES) to evaluate patient scheduling strategies.
- ▶ **Key Finding:** Dynamic scheduling (Slack-Based Policy) significantly reduces patient wait times compared to static rules (First-Come, First-Served).

Problem Statement & Scope

The Problem: Hospital outpatient departments struggle to balance patient satisfaction (low wait times) with financial constraints (minimizing idle doctors) during unpredictable peak hours.

Our Scope & Motive (Methodological Extension):

- ▶ **Optimize Resource Allocation:** We will extend the paper's model to find the exact staffing threshold needed to minimize wait times without overstaffing.
- ▶ **Enhance Stability:** We will test a novel dynamic priority rule to prevent system bottlenecks and ensure stability under high patient loads.
- ▶ **Improve Computational Efficiency:** Exact mathematical derivation for hospital queues is computationally intractable. By utilizing Discrete Event Simulation (DES), we can achieve a highly efficient, robust computational platform for this problem.

Expected Methodology

1. Stochastic Modeling (Random Number Generation)

We will implement RNG to generate stochastic patient inter-arrival times (modeling a Poisson process) and doctor service times (modeling exponential distributions) based on empirical dataset.

2. Discrete Event Simulation (DES) Architecture

We will build an event-driven algorithmic loop that chronologically processes core events (*Patient Arrival*, *Service Start*, and *Discharge*) to calculate state changes rather than ticking second-by-second.

3. Grid Search Optimization

We will iteratively run the simulation to perform a grid search, identifying the optimal parameters to guarantee maximum service level compliance.